Unit title: Working with IT Software – Spreadsheet and Database
Unit code: F1FJ 12

Unit Outline
This Unit is designed to introduce complex features of spreadsheet and database application software. The candidate will gain practical experience in the use of complex features in these types of software. The Unit will also provide candidates with information regarding the selection of appropriate software for specific tasks.
The Unit is appropriate for any candidate who wishes to enhance their IT skills in spreadsheet and database application software.

Outcomes
Learners who complete this Unit will be able to:

Outcome 1
Identify complex computer application software functions.
1.1 Identify complex functions of a spreadsheet application package.
1.2 Identify complex functions of a database application package.

Outcome 2
Explain issues relating to data and information.
2.1 Explain the relevance of information produced by spreadsheets and database for different audiences.
2.2 Identify issues relating to data integrity and accuracy.

Outcome 3
Use complex computer application software functions.
3.1 Use complex functions of a spreadsheet application package.
3.2 Use complex functions of a database application package.

Outcome 4
Demonstrate and review the use of complex application software functions to produce solutions to specified tasks.
4.1 Select an application package for each specified task.
4.2 Justify choice of application package for the specified tasks.
4.3 Select appropriate complex information from a variety of sources.
4.4 Produce a solution to specified tasks.

4.5 Review the use of IT resources to produce a solution for the specified tasks and changes to improve efficiency when carrying out similar tasks in the future.

Guidance on the content

The overall aim of this Unit is to introduce the candidate to the complex features of spreadsheet and database application software. On completion of the Unit candidates should also be able to make an informed choice as to which software to use for a particular task. The Unit is designed to be task based and it is anticipated that most of the time will be spent on practical work, although some discussion work will be required.

Outcomes 1 and 3

This Outcome covers the practical use of a range of application software. Exercises should be presented for the candidate to work through in order that they gain the practical skills necessary to demonstrate proficiency in the complex skills outlined below.

It is expected that candidates undertaking this Unit will possess a working knowledge of the software application packages covered within this Unit.

- Spreadsheets:

  This section should allow the candidate to produce spreadsheets for analysing and interpreting complex data such as budgets and annual accounts. Candidates should be able to use appropriate tools to predict trends and understand how to produce information that communicates effectively by structuring the content to take account of different contexts and audience needs. Candidates should be able to use tools to automate routine tasks.

- Database:

  This section should allow the candidate to use multiple table databases. One-to-one and one-to-many relationships should be covered.

  Candidates, at this level, should be aware of integration between packages. A word processing package could be used to illustrate how the data in a spreadsheet and database can be used in an integrated document. This could take the form of a mail merge or incorporating a database and spreadsheet into a document.

Outcome 2

This Outcome deals with issues relating to information produce by different software application packages and the appropriateness of the information produced for different

audiences. The candidate will also be introduced to issues relating to data integrity and the accuracy of information.

Outcome 4

This Outcome deals with the candidate's ability to select an appropriate application package for a specified task. Candidates should be given a task and asked to say which of the two types of application software covered in Outcome 1 is most suitable for the specified task.

Two specified tasks which are designed for separate application packages are required for Outcome 4. These tasks must be completed by the candidate to achieve the Performance Criteria for Outcome 4.

The tasks will be aimed at different audiences, ie requiring different levels of complexity, and the candidate must describe how the different audience requirements will be met by the candidate's use of the application packages taking into account:

- accuracy and effectiveness of information
- structured content to take account of different contexts
- structured content to take account of different audiences
- production of information in the right place at the right time

On completion of the portfolio produced in Outcome 4 the candidate will provide a short report detailing the following:

- how the candidate can share the knowledge and skills gained in this Unit to help others
- using the feedback to determine what steps can be taken to improve any weaknesses
- how the impact of the work produced can have an impact on others

Candidates should be able to work out how to use IT effectively for technically complex tasks and purposes, taking account of their own and others skills and capabilities and the needs of the organisation. This will involve working out how to use IT for technically complex tasks (eg, doing a cost benefit analysis, reporting the results of a survey about clients' needs and preferences).

Candidates should be able to identify an appropriate application package for a specific task and explain why and how using the IT system and software was an appropriate way of carrying out the task.

Having completed the tasks for Outcome 4 candidates should then review the appropriateness and effectiveness of their decisions and actions taken about the choice and use of software tools and techniques, in relation to the task or purpose involved. This will involve the candidate identifying changes that could make similar tasks and purposes easier or more successful. Candidates should be given feedback from other people regarding their use of IT and should evaluate this feedback, giving and detailing steps that could be taken to improve any weaknesses. Candidates are required to comment on the impact of their own use of IT in their review.

In their review, candidates should demonstrate ability to use a wide range of correct terms for hardware and software. Candidates should be able to provide a simple explanation of IT terms that others may find difficult to understand.

This Outcome deals with the candidate's ability to produce a solution to a given task after they have decided which application package is most suitable for the task. This Outcome is designed to allow candidates to demonstrate skills from both Outcomes 1, 2 and 3. In Outcome 4 the candidate has gained the knowledge to make an informed decision and the candidate should apply the skills that he/she has acquired in Outcome 1 and 3 to produce a solution to the specified task.

Opportunities For Core Skill Development

This Unit involves candidates:

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- Using a wide range of IT and carrying out searches to extract and present relevant information which may provide an opportunity to gather evidence towards aspects of the IT Core Skill.
- Making selective use of IT and evaluating their choice which may which may provide an opportunity to gather evidence towards aspects of the Problem Solving Core Skill.
- Reporting a number of aspects of the assessment which may provide an opportunity to gather evidence towards aspects of the Communications Core Skill.